piRover Builds with K2

Creating a Remote Connection Rev 3.0

Overview:

In this activity, you will connect to your Raspberry Pi and complete configuration steps so that you will have remote access to your piRover. You'll download and install the required software on your computer system and make your first remote connection into the system.

Prerequisites:

Prior to beginning the instruction provided in this lesson you must have completed the following:

1. piRover build and test

Performance Outcomes:

1. Connect to a Wi-Fi access point.
2. Edit a configuration.
3. Modify the Wi-Fi access point name and authentication.

Resources:

1. 2024 piRover kit with K2 microSD card image

Materials:

1. Wi-Fi access point
2. Laptop or Workstation with Ethernet network connection
3. Ethernet cable (provided in kit)
4. Ethernet cable adapter (requirement discussed in week 1)
5. VNC Viewer
6. Bonjour Print Services for Windows

Directions:

1. A typical setup for a Raspberry Pi single-board computer is shown in Figure 1. Note the USB-C power connection in addition to the USB Dongle providing keyboard and mouse connectivity. A small HDMI monitor is displaying the Pi desktop and requires the Micro-HDMI cable included in your kit.

© 2024, Keith E. Kelly. All rights reserved.

piRover Builds with K2

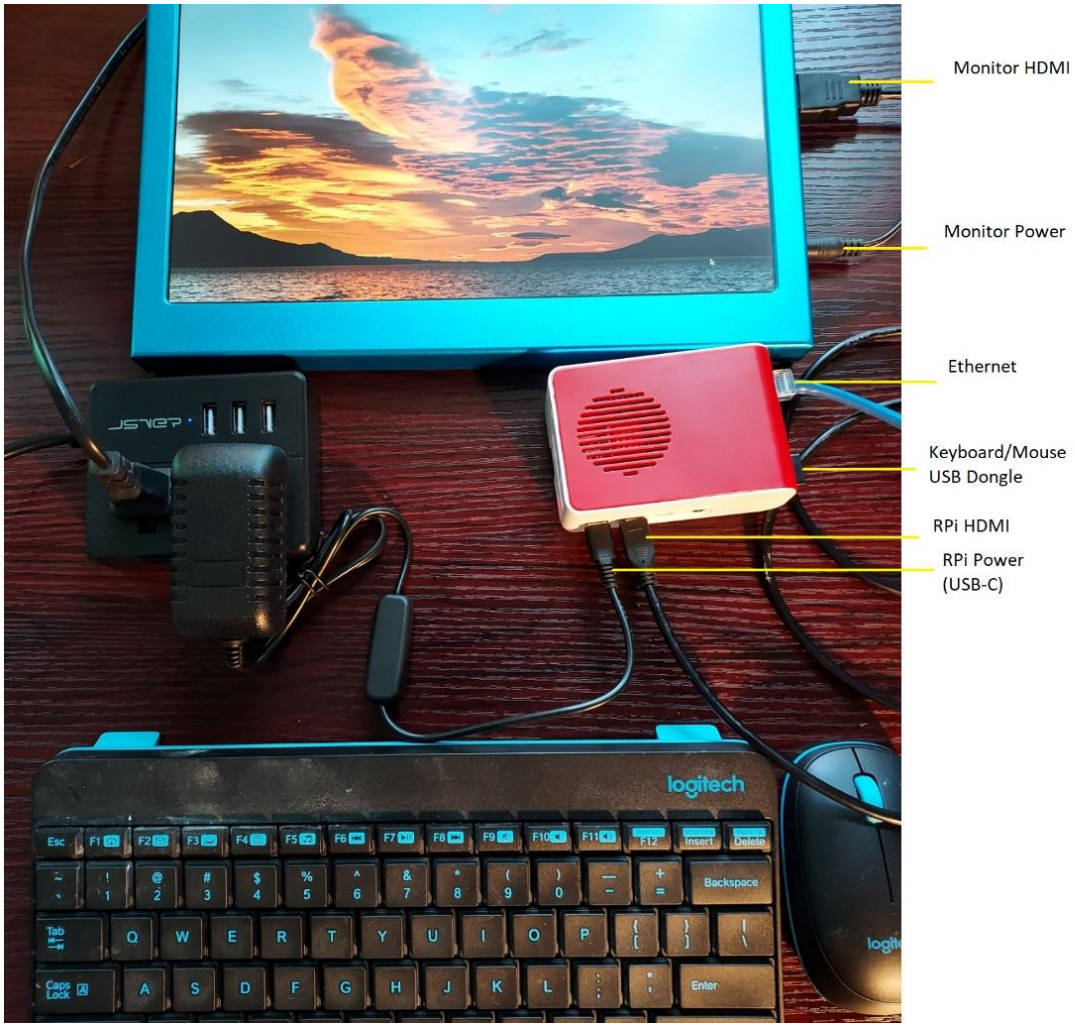


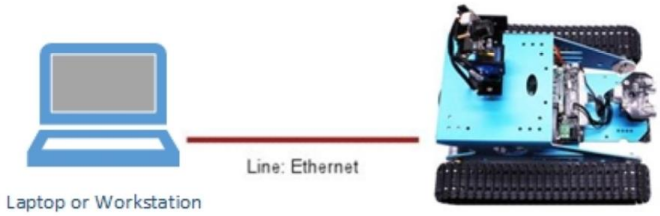
Figure 1 - RPi Single Board Computer Setup

2. In this class, you are connecting to the Raspberry Pi integrated into your piRover system. Typically, you will not have a monitor, keyboard, or mouse directly connected to the Pi, but these components may be required if you are having issues connecting. Normally you will connect using a remote desktop service (VNC) running on the Pi. This is often referred to as "headless mode."

3. In this activity, you make your first connection to the RPi using the Ethernet cable provided. Eventually you may be able to connect using Wi-Fi connectivity and the Ethernet connection is not required.

4. The first step in making the remote connection is to connect the piRover to your computer using the Ethernet cable provided.

© 2024, Keith E. Kelly. All rights reserved.



5. Recall that if your computer does not have an Ethernet port, a USB to Ethernet adapter is required like the one shown below. This requirement was discussed during the first week of class.

Figure 2 Ethernet to USB adapter

6. You'll be creating a peer-to-peer network connection between your workstation and the Raspberry Pi. If you are using a Windows PC an additional software driver is required to make this type of connection. If you are on a PC, navigate to this Bonjour Print Services for Windows link. Download and install the driver. This step is not required if you are using an NMC workstation in a PS lab.

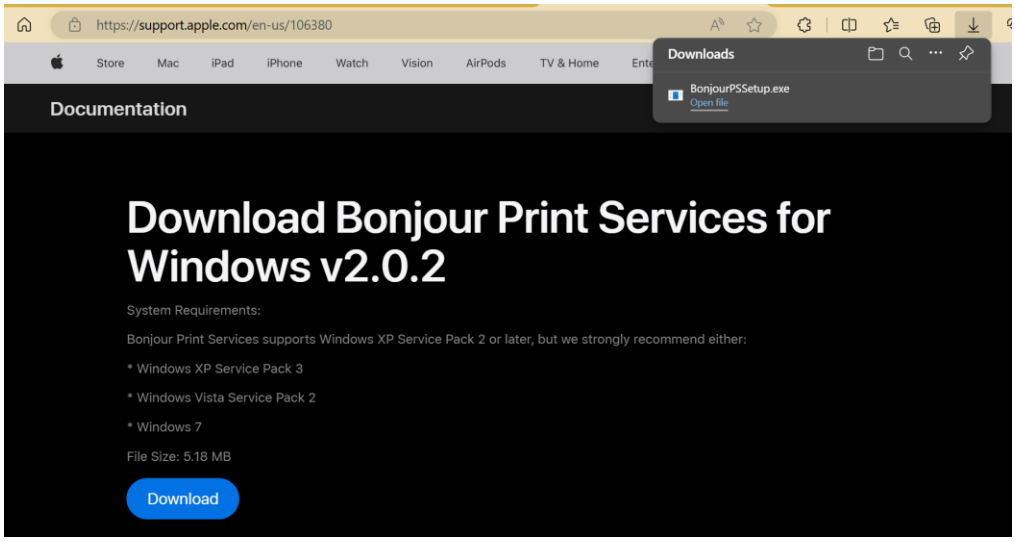


Figure 3 Bonjour Print Services for Windows

© 2024, Keith E. Kelly. All rights reserved.

1. You'll use VNC remote desktop software to provide the remote connection. The RPi system provided for you has the VNC server feature enabled as shown below, but you must install the VNC client on your workstation. Again, this step is not required if you are using an NMC workstation in a PS lab.

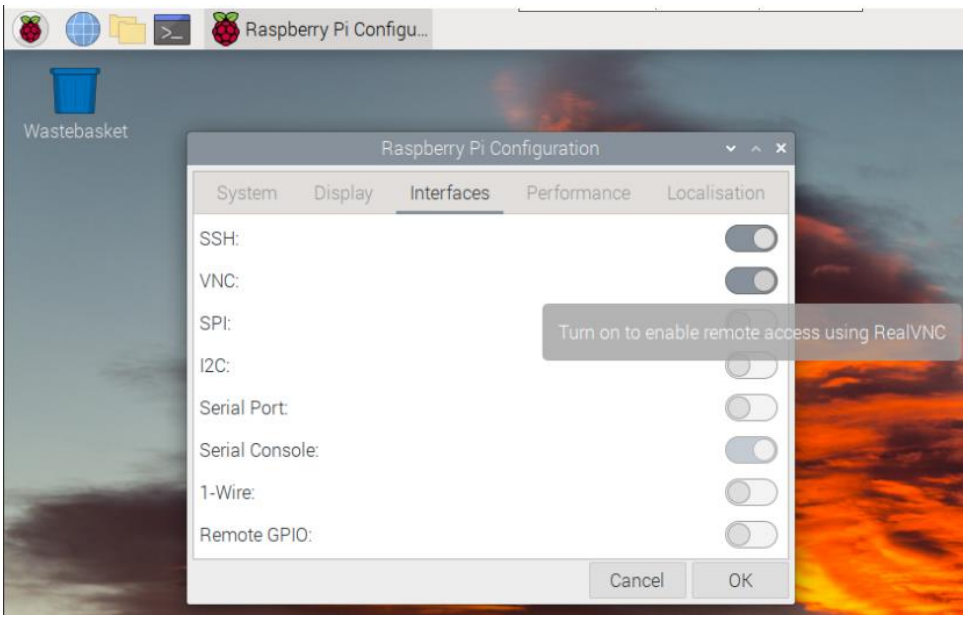


Figure 4 Remote Desktop showing VNC Server Enabled

7. Open the VNC Viewer link on your workstation. Download and install the VNC Viewer client software. **Just install the viewer software.** The software is free, and you DO NOT need to sign up for additional software or services such as VNC Connect.

8. You are now ready to connect.
    a. Turn the power on to the piRover. Note the red LED indicating that power is applied and the flashing green LED light indicating activity.

    b. Allow time for the piRover to complete its boot process. The Yahboom software loads by default and you will hear four beeps indicating the Yahboom service on the piRover is ready.

9. Open VNC Viewer on your workstation. A typical application window is shown on the following page. The application window is blank at

© 2024, Keith E. Kelly. All rights reserved.

this time due to no prior remote connection. Icons will appear here once you successfully connect.

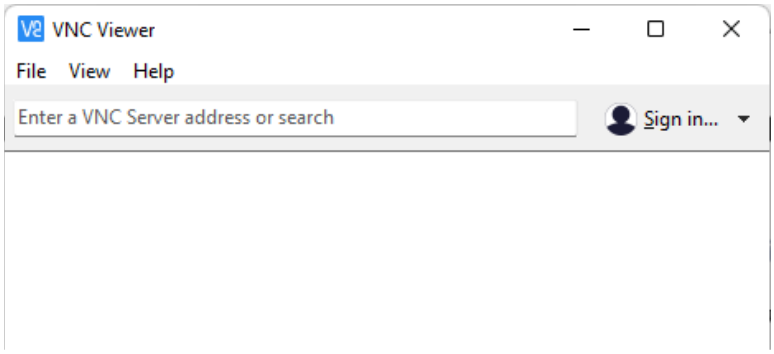


Figure 5 VNC Client Window

10. Enter the VNC Server address below. The server name (or hostname) is **piRover**. The **.local** is appended to the end due to the peer-to-peer network connection that is created over the Ethernet.

    `piRover.local`

11. You must provide credentials to create the remote connection. Enter **pi** as the user and **nmc_ram** as the default password. You will continue to use pi as the user throughout the course. You will change the default password once you are connected.

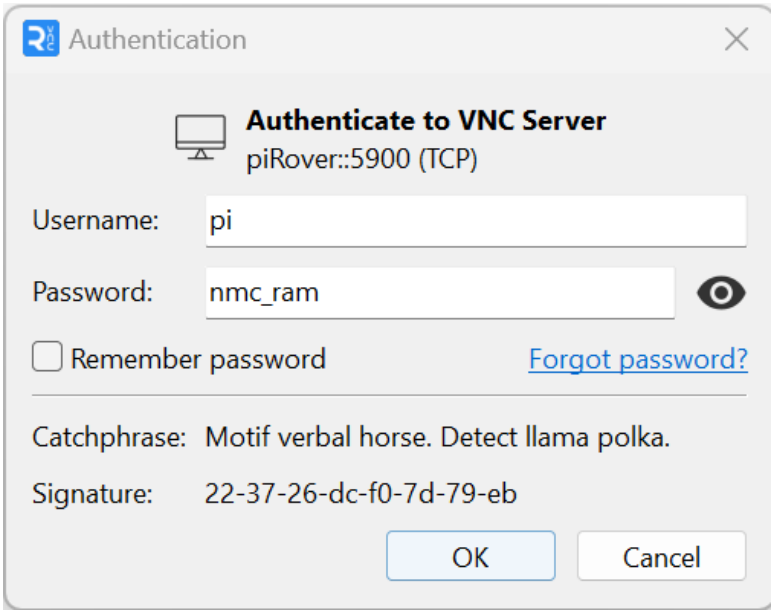


Figure 6 VNC Authentication

© 2024, Keith E. Kelly. All rights reserved.

12. Click OK. A remote desktop is displayed.

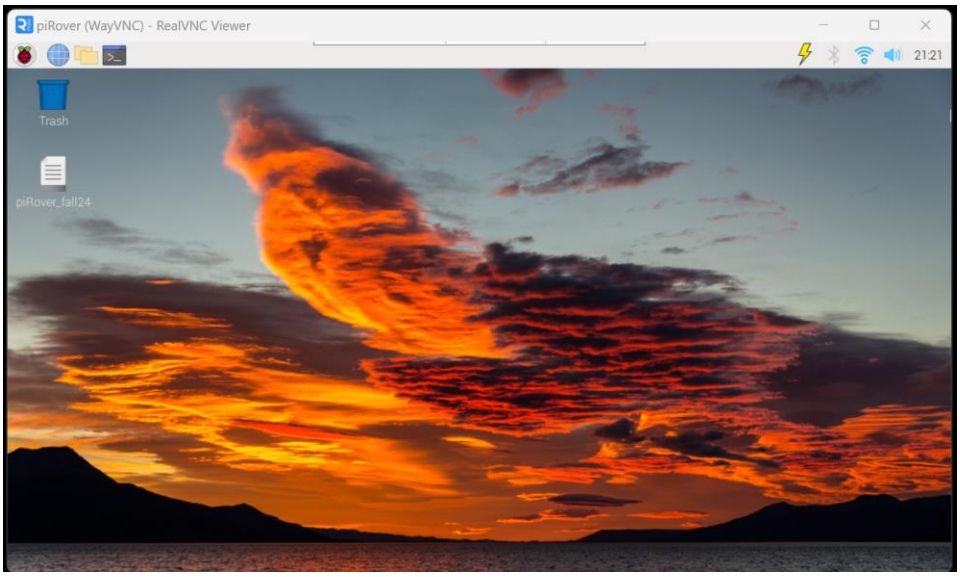


Figure 7 RPi Remote Desktop using VNC Viewer

13. Change the initial ***nmc_ram*** password to secure your system. Click the Raspberry in the top left corner and then click Preferences and then Raspberry Pi Configuration as shown below.

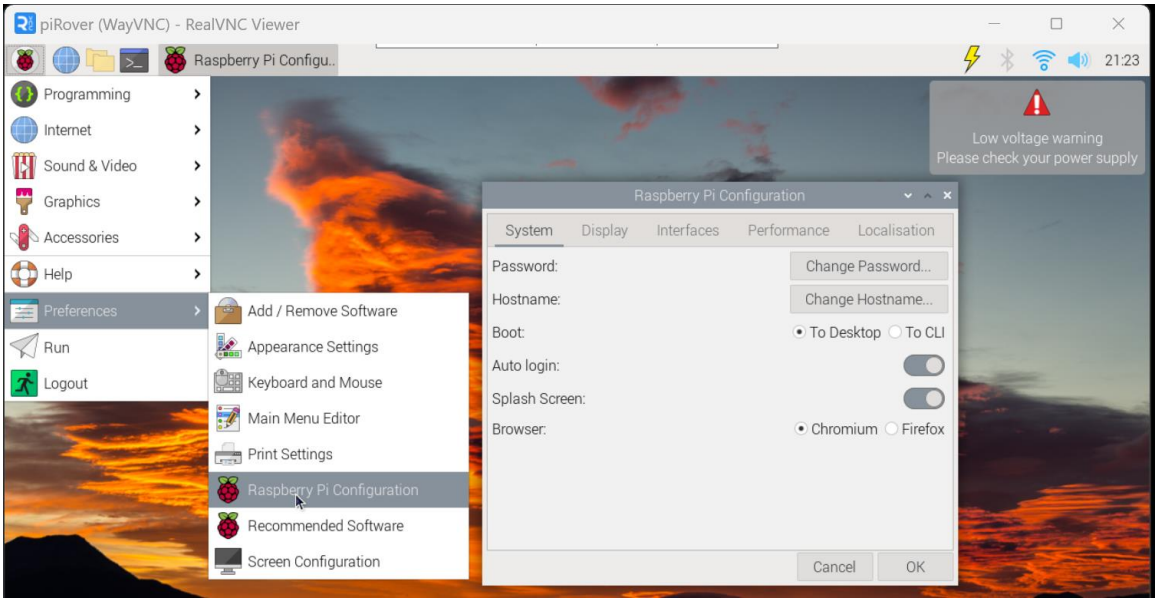


Figure 8 RPi Menu and Configuration

14. Click the Change Password button and enter your new password. Confirming the password will activate the OK button. Click OK and then close the configuration window. **NOTE:** *You must remember your password!* There is no administrative override or ability to reset.

© 2024, Keith E. Kelly. All rights reserved.

15. If you forget your password, you'll need to start with a new image of the operating system, reconfigure, and then restore any project work from your cloud storage.
16. Now that the password is updated. Restart the piRover using the menu as shown below.

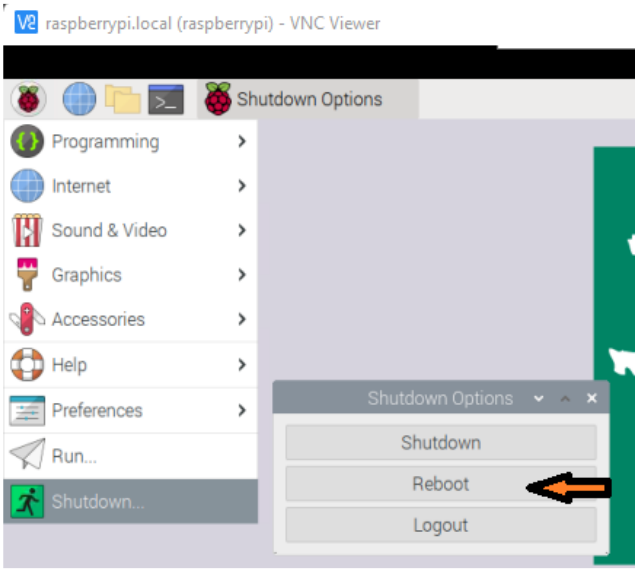


Figure 9 Reboot using the RPi Menu

17. The VNC view will lose the connection and attempt to reconnect.

18. Once the piRover boots, the Authentication screen will be displayed again. Enter your new password, click the *Remember password* check box, and click OK. The remote desktop will display.

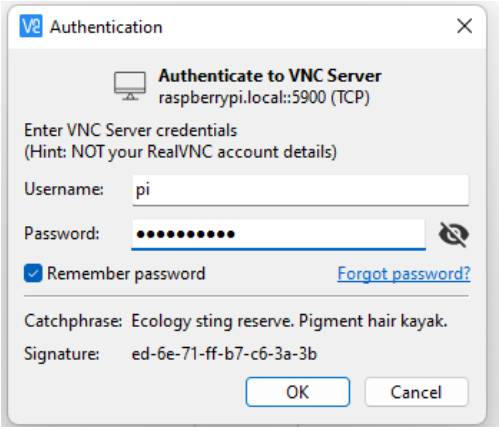


Figure 10 VNC Authentication with Remember Checkbox Ticked

19. The next step is to configure the piRover's Wi-Fi connection. Open the remote desktop and locate the Wi-Fi icon in the upper righthand location on the desktop.

© 2024, Keith E. Kelly. All rights reserved.

20. Note this double-arrow icon shown on the right side of the taskbar. This symbol indicates that there is not a current Wi-Fi connection.

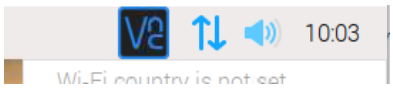


Figure 11 Taskbar showing no Wi-Fi connectivity

21. Click on this icon to select the Wi-Fi access point from the available list as shown in the image below. The instructor will review access options if you are on NMC's campus. If you are connecting to your home access point or smartphone hotspot, you will need the required password or pass phrase.

Figure 12 Taskbar showing sample AP connections

22. Connecting to a secure Wi-Fi access point (AP) will cause the following password prompt to be displayed. Enter the required password and click the Connect button.

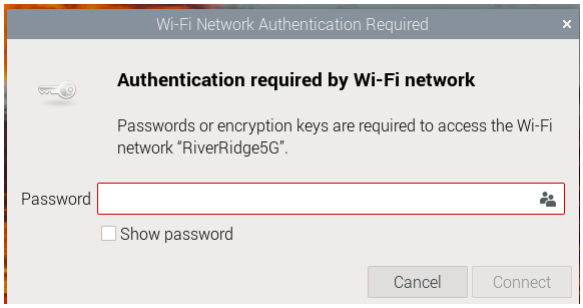


Figure 13 Wi-Fi authentication prompt

23. The icon on the taskbar will update to a typical Wi-Fi icon as shown below. Hovering over this icon will display details of your connection.

Figure 14 Wi-Fi connected with IP details shown

© 2024, Keith E. Kelly. All rights reserved.

24. Verify that you have an Internet connection on the piRover by launching the browser using the icon on the taskbar and navigating to Google.com. The Google search page should load.

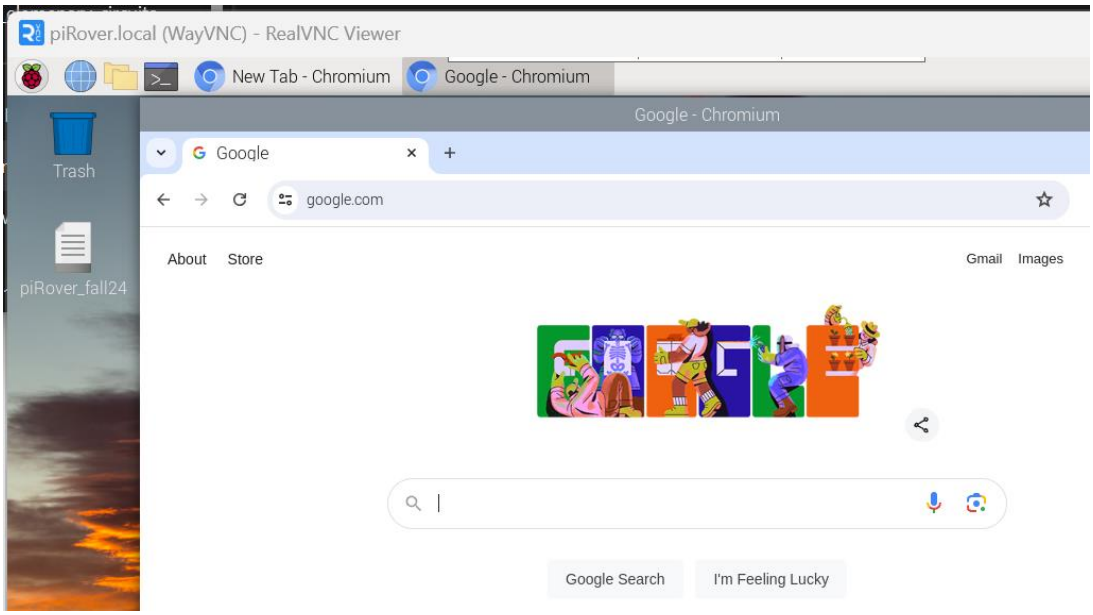


Figure 15 Browser with Google page loaded

25. A more common way for a developer to check for an Internet connection is to "ping" a server. Open a terminal window and enter the following command. This "pings" the Google name space server at 8.8.8.8. If your connection is good, you will see replies as shown below. Close the window or use Ctrl+C to stop.

ping 8.8.8.8

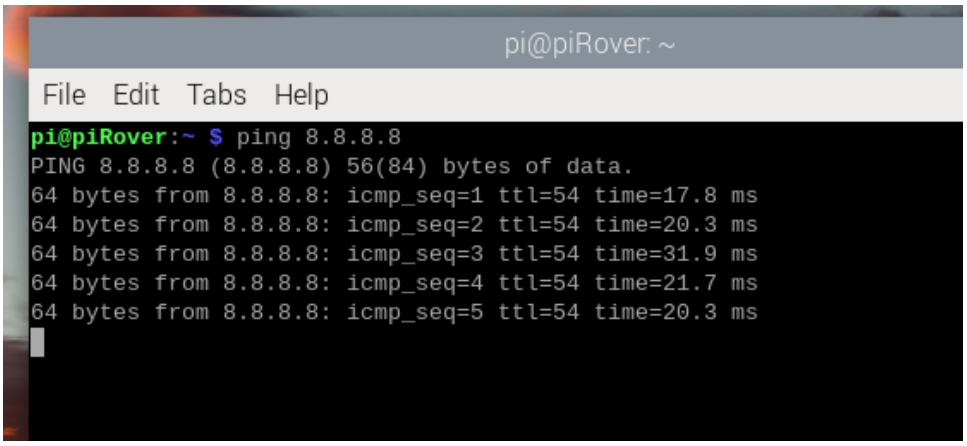


Figure 16 Ping command showing replies

© 2024, Keith E. Kelly. All rights reserved.

piRover Builds with K2

Assessment:

Your instructor will discuss any assessment action due at this time. You must be able to remotely connect to the piRover. All future piRover class activities are dependent on this remote connection.

© 2024, Keith E. Kelly. All rights reserved.